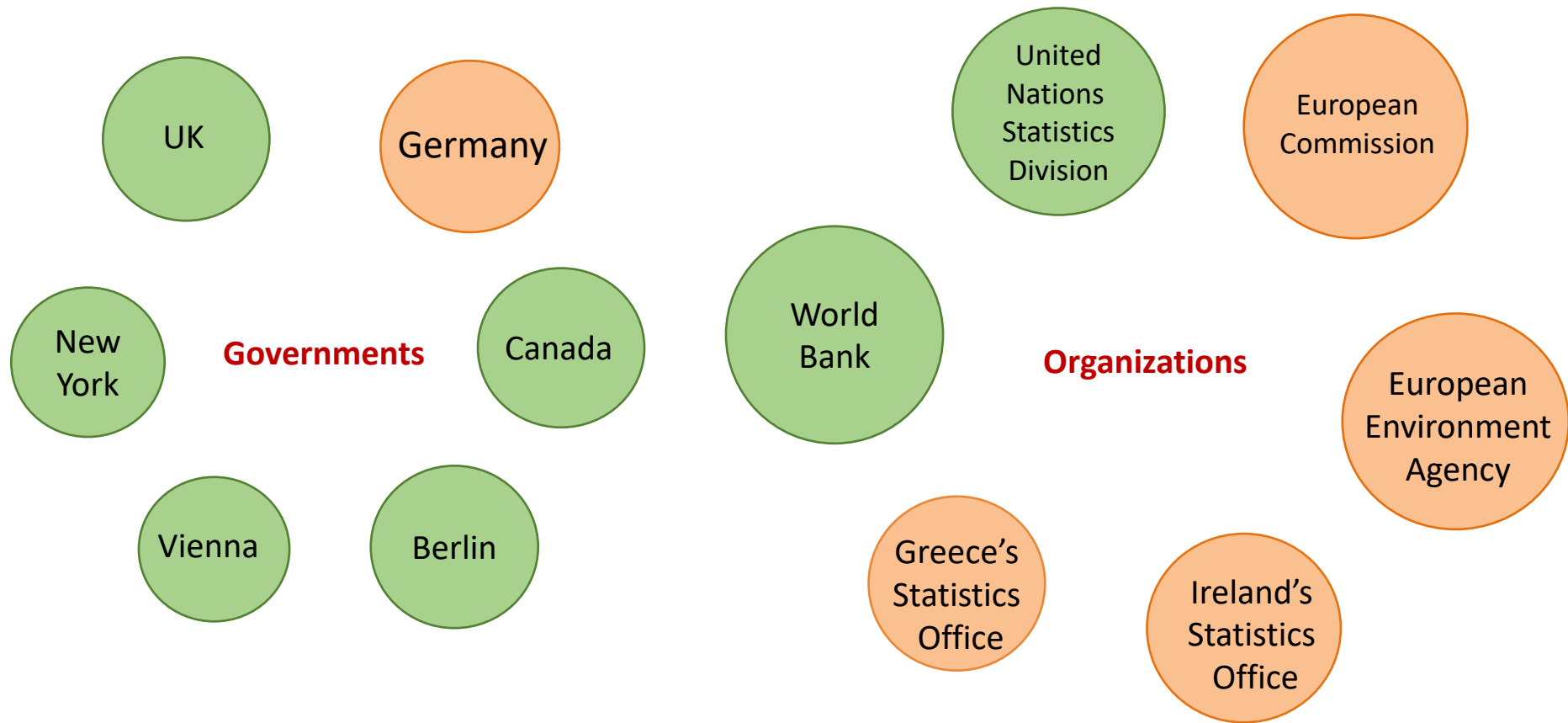




ĐẠI HỌC BÁCH KHOA HÀ NỘI
VIỆN CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG

Semantic Exploration and Integration of Statistical Data



Statistical data is published by various governments and organizations!

?

UK

Germany

?

United Nations
Statistics
Division

?

European
Commission

?

World
Bank

Governments

New
York

Canada

Organizations

European
Environment
Agency

Vienna

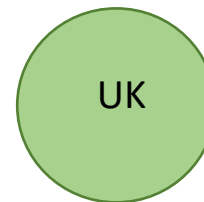
Berlin

Greece's
Statistics
Office

Ireland's
Statistics
Office

What is the population
of the UK?



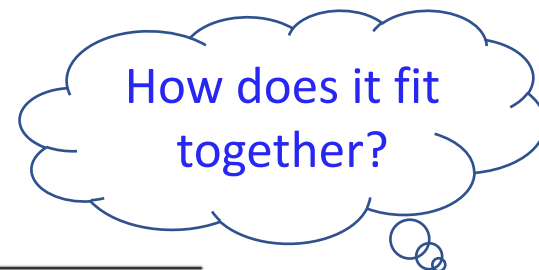


```
{
  "indicator": {
    "id": "SP.POP.TOTL",
    "value": "Population,_total"
  },
  "country": {
    "id": "GB",
    "value": "United_Kingdom"
  },
  "value": "64128226",
  "decimal": "0",
  "date": "2013"
}
```



Heterogeneity of formats,
encodings, scales,
structures, access mechanisms

Mid-Year	Mid-Year Population (millions)	Annual Percentage Change
2011	63.3	0.84
2012	63.7	0.66
2013	64.1	0.63

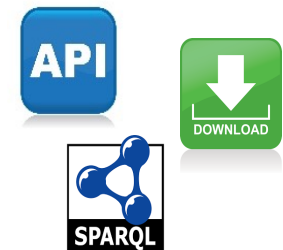
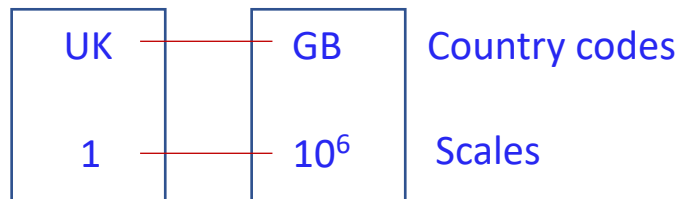
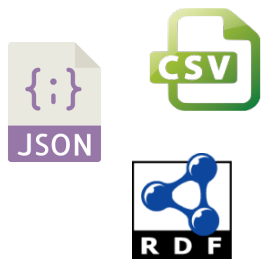


sdmxid: freq	sdmxid: timePeriod	sdmxid: refArea	sd- mxid:age	sdmxid: sex	sdmxm: obsValue
sdmx-code:freqA	2013	geo:UK	ag:TOTAL	sdmxcode: sex-T	63,905,297
sdmx-code:freqA	2013	geo:UK	ag:Y_LT15	sdmxcode: sex-T	11,260,549
sdmx-code:freqA	2013	geo:UK	ag:Y15-64	sdmxcode: sex-F	20,917,257



Motivation & Challenges

- Motivation: facilitate exploration and integration of heterogeneous statistical data sets
- Main challenges
 - Syntactic heterogeneity
 - Semantic heterogeneity
 - Access heterogeneity



Research Gaps

- Statistical data **integration**

“The integration of data cubes is still an unexploited research topic”

Karamanou et al., *Linked Data Cubes: Research results so far*

- Statistical data **exploration**

- Focusing on visualization of individual data sets [1-7]
- Identification of relatable data set is an open problem

Research Questions

How can users be enabled to explore and integrate heterogenous statistical data sources?

- RQ1.** How can we address statistical data heterogeneity in terms of formats?
- RQ2.** How can we establish interconnections between statistical data sets?
- RQ3.** How can we provide uniform access to heterogenous statistical data sets?

Syntactic Heterogeneity

RQ1. How can we address statistical data heterogeneity in terms of formats?

- **RDF**: standard for data representation [9, 10]

	In advance transformation	Query time transformation
Mechanism	Transforms data into RDF then stores data in endpoints [11-15]	Transforms data into RDF when requested by users
Options	Mapping languages [11, 12], ETL tools [13- 15]	Mapping languages
Advantages	Manages data sets easily	Provides up-to-date data Requires small volume for mapping storage
Disadvantages	Increases data volume Faces with out-of-date data	Increases query answering time

- Approach
 - Uses mapping language for data transformation at query time
 - Creates a cache to reduce query answering time

Mapping Languages

W3C

	CSV	Spreadsheet	JSON	XML	Database
XLWrap [16]	✓				
M ² [17]		✓			
XSPARQL [18]				✓	
D2RMap [19]					✓
R2RML [20]					✓
M2RML [21]					✓
RML [22]	✓		✓	✓	

An overview of existing mapping languages

- Choose RML for data transformation
- Two extensions for the RML's processor
 - Support for spreadsheet formats
 - Support for variables in mappings

RML mapping for the World Bank

<http://statspace.linkedwidgets.org/mapping/wb.ttl&indicator=NY.GDP.PCAP.CD&refArea=AT>

<#Variables>

rml:defaultValue

[rml:varName "**indicator**"; rr:constant "SP.POP.TOTL"],
[rml:varName "**refArea**"; rr:constant "all"].

<#Observation>

rml:logicalSource [

rml:source

"http://api.worldbank.org/countries/{**refArea**}/indicators/{**indicator**?
format=json&page=1&per_page=15000";

rml:referenceFormulation ql:JSONPath;

rml:iterator "\$[1].*"

];

AT

NY.GDP.PCAP.CD

Description of the input data set

rr:subjectMap[

rr:class qb:Observation;

rr:template "http://statspace.linkedwidgets.org/dataset/WorldBank-
{indicator.id}/Obs-{country.id}-{date}";

rr:termType rr:IRI

];

Template to coin subjects

rr:predicateObjectMap [

rr:predicate sdmxd:refArea;

(Subject, Predicate, Object)

rr:objectMap [

rr:template

"http://statspace.linkedwidgets.org/codelist/cl_area/{country.value}";

rr:termType rr:IRI

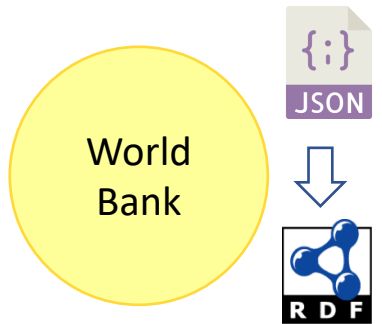
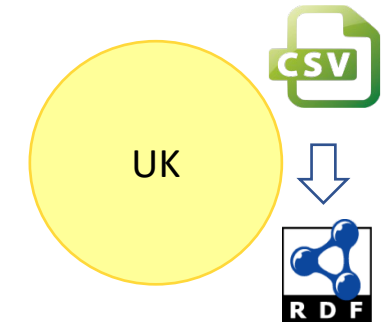
Template to coin objects

];



Now, data is syntactically homogenous (RDF)

Heterogeneity of URIs and scales



uk:country	uk:year	uk:value	uk:unit
uk:UnitedKingdom	uk:2013	64.1	uk:Million

wb:country	wb:year	wb:value	wb:unit
wb:GB	wb:2013	64128226	wb:AbsoluteScale

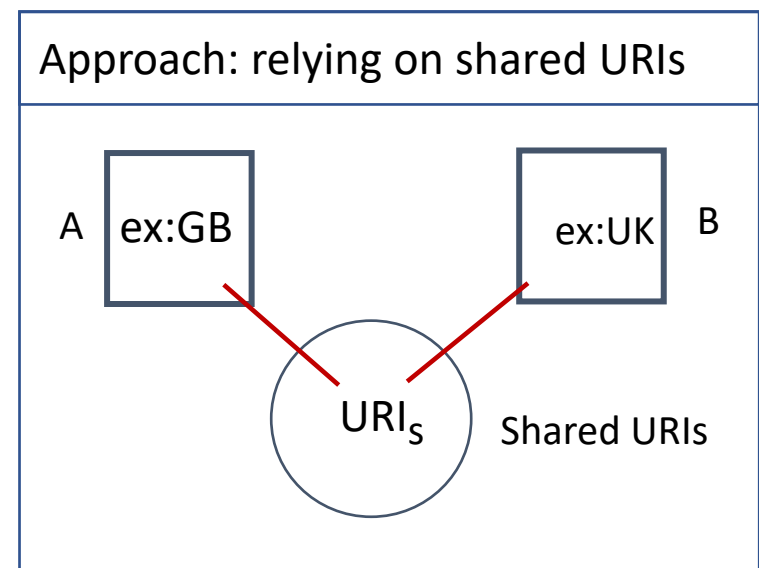
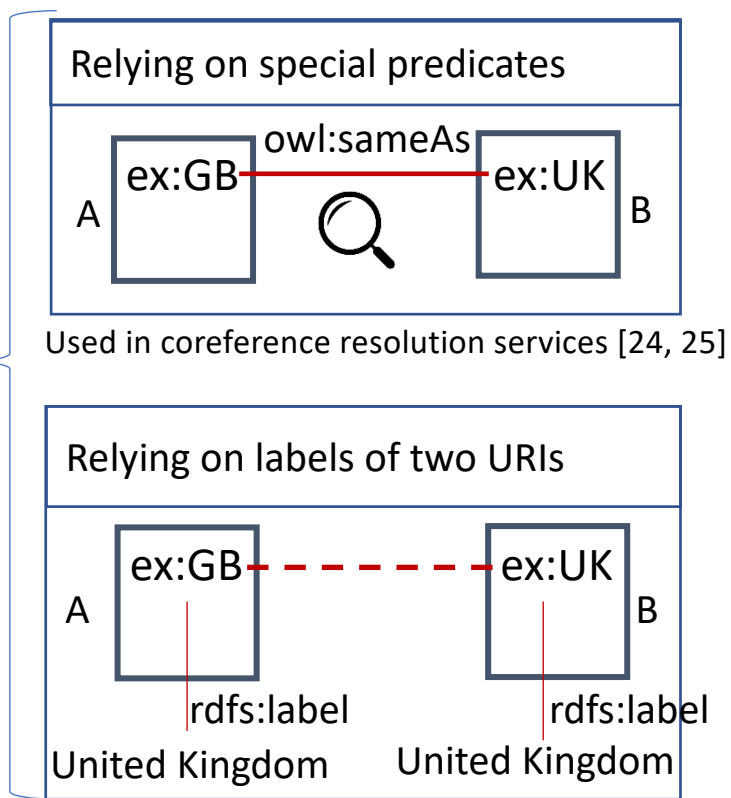
sdmxd:freq	sdmxd:timePeriod	sdmxd:refArea	sd-mxd:age	sdmxd:sex	sdmxm:obsValue
sdmx-code:freqA	2013	geo:UK	ag:TOTAL	sdmxcode:sex-T	63,905,297
sdmx-code:freqA	2013	geo:UK	ag:Y_LT15	sdmxcode:sex-T	11,260,549
sdmx-code:freqA	2013	geo:UK	ag:Y15-64	sdmxcode:sex-F	20,917,257

Semantic Heterogeneity

RQ2. How can we establish interconnections between statistical data sets?

- Basics for data integration: identify equivalent entities

establish
direct
relationships



A set of shared URIs for linking facilitates query rewriting and result rewriting

Shared URIs – Example

- Geographical areas
 - Pattern: *Country/Administrative Area Level 1/.../Administrative Area Level n*
 - Example: Austria/Vienna/Vienna/Floridsdorf
- Temporal values
 - Patterns: rely on the UK time reference service
 - Example:
 - Year: <http://reference.data.gov.uk/id/gregorian-year/{year}>
 - Month: <http://reference.data.gov.uk/id/gregorian-month/{month}>

Shared URIs

http://statspace.linkedwidgets.org/codelist/cl_area/UnitedKingdom

Geographical areas

SDMX vocabulary

`sdmx:refArea,`
`sdmx:refPeriod`

Recommended codes of ISO standards

Shared URIs

UK's time reference service

<http://reference.data.gov.uk/id/gregorian-month/2017-08>

Subjects of data sets

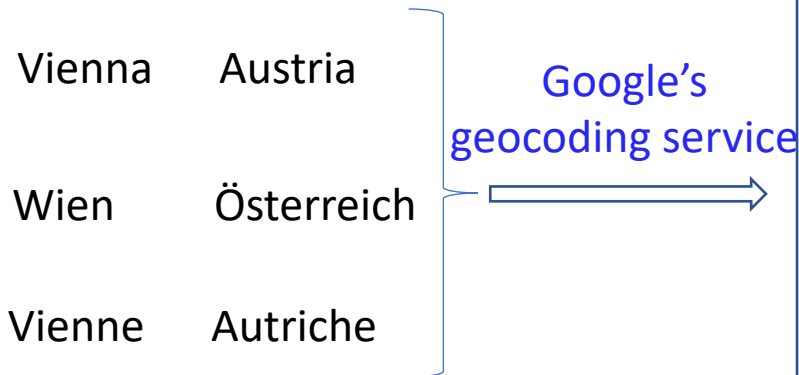
Units

http://statspace.linkedwidgets.org/codelist/cl_unitMeasure/P6

<http://statspace.linkedwidgets.org/codelist/SP.POP.TOTL>

Spatial Dimension Mapping Algorithm

- Spatial dimension: maps different URIs, such as *ex:country*, *ex:area*, etc. to *sdmxd:refArea*
- Spatial values: uses Google's geocoding service for mapping discovery



```
"results" : [
  {
    "address_components" : [
      {
        "long_name" : "Vienna",
        "short_name" : "Vienna",
        "types" : [ "locality", "political" ]
      },
      {
        "long_name" : "Vienna",
        "short_name" : "Vienna",
        "types": [ "administrative_area_level_1", "political" ]
      },
      {
        "long_name" : "Austria",
        "short_name" : "AT",
        "types" : [ "country", "political" ]
      }
    ],
    "formatted_address" : "Vienna, Austria",
    "geometry" : {
      "location" : {
        "lat" : 48.2081743,
        "lng" : 16.3738189
      }
    }
  }
]
```

Now, RDF data sets are linked!



UK

?o uk:year uk:2013



World
Bank

?o wb:year wb:2013



European
Environment
Agency

?o sdmx:timePeriod 2013

Heterogeneity of access

<http://reference.data.gov.uk/id/gregorian-year/2013>

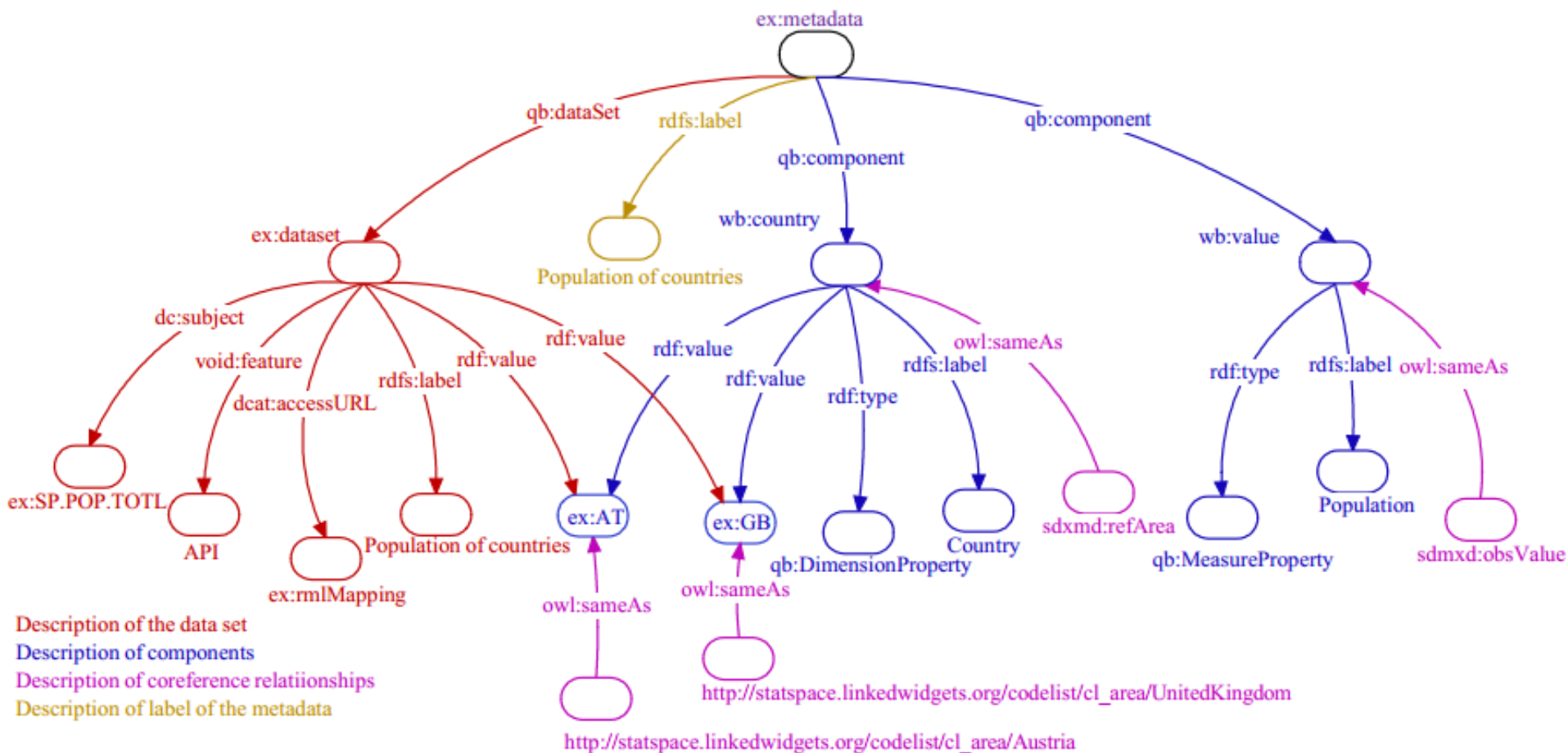
Is there a way to provide
uniform access?

Uniform Access

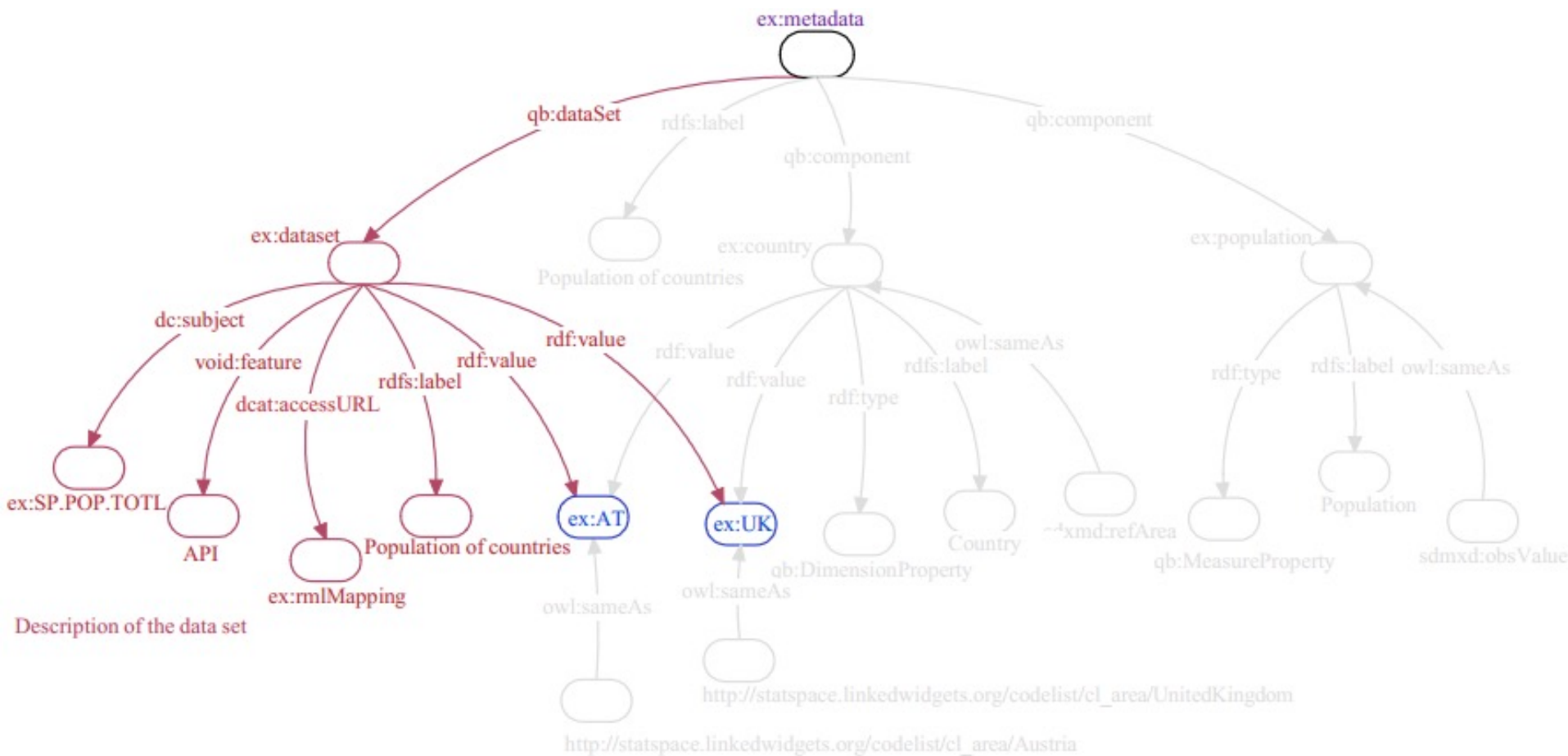
RQ3. How can we provide uniform access to heterogeneous statistical data sets?

- No related work on statistical data domain!
- Approach
 - Metadata description for each statistical data set
 - Information about data structure, access mechanism, and equivalent relationships
 - Standardized conceptual layer over each data set
 - Mediator
 - Query rewriting and result rewriting
 - Single point of access

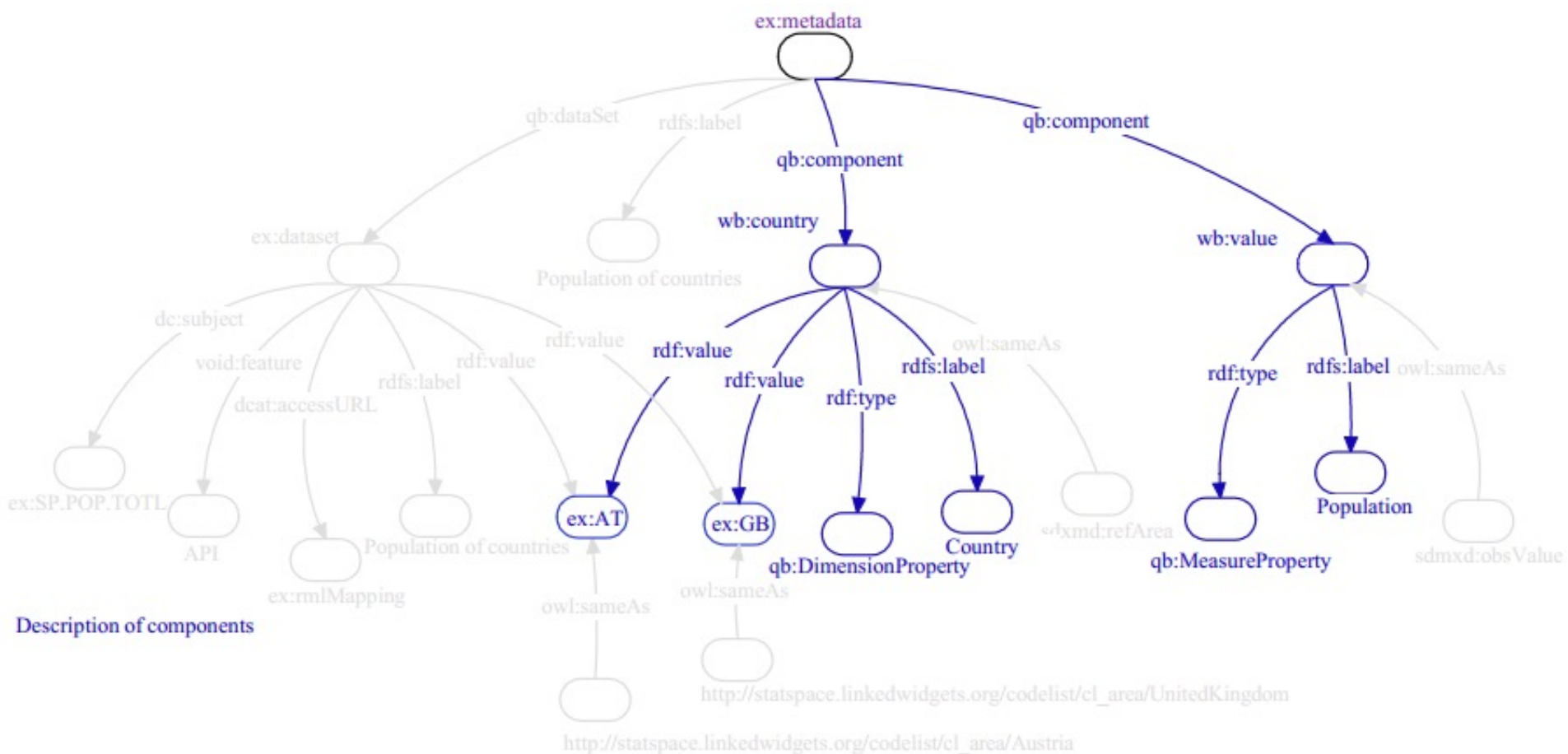
Metadata Description - Example



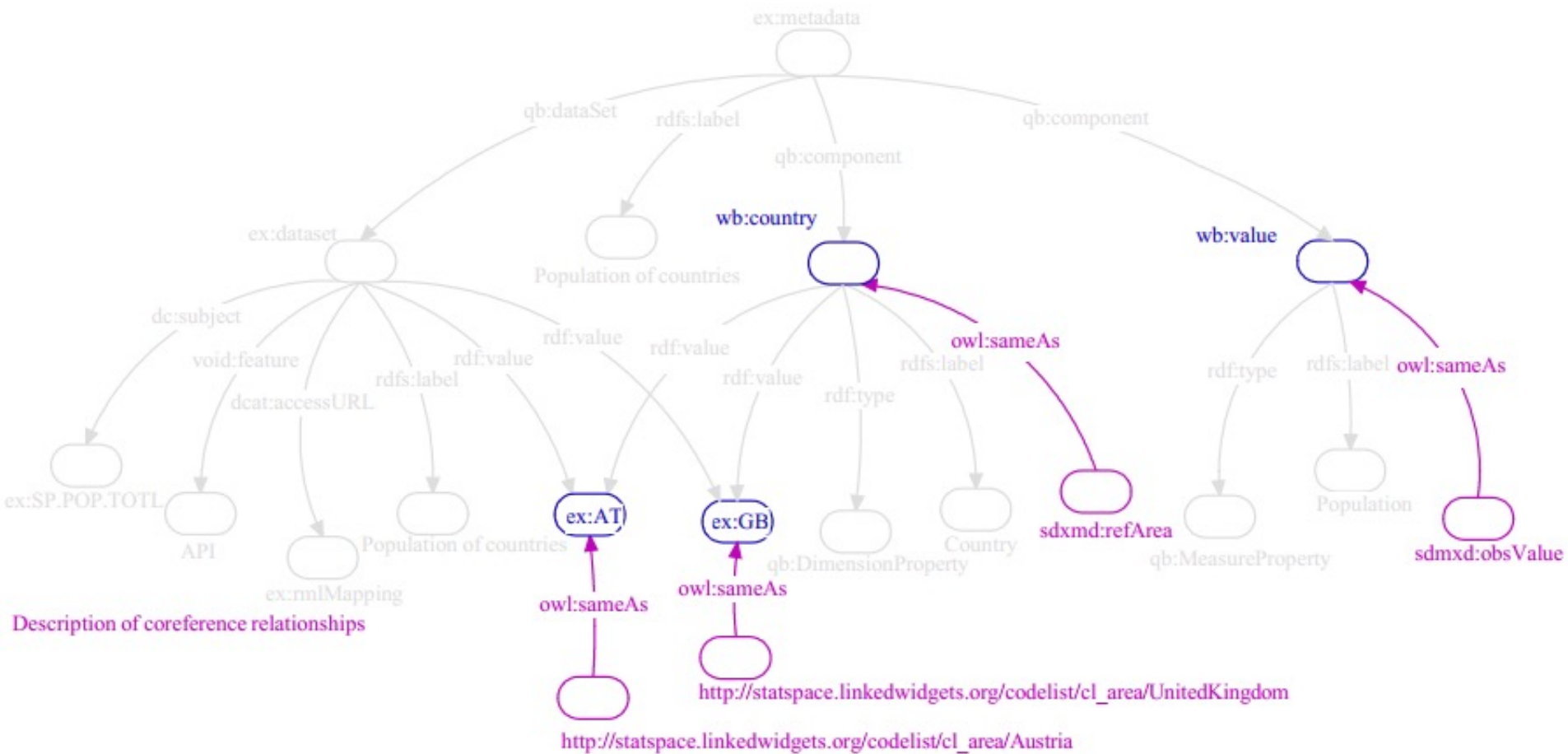
Metadata Description – Description of the data set



Metadata Description – Description of components

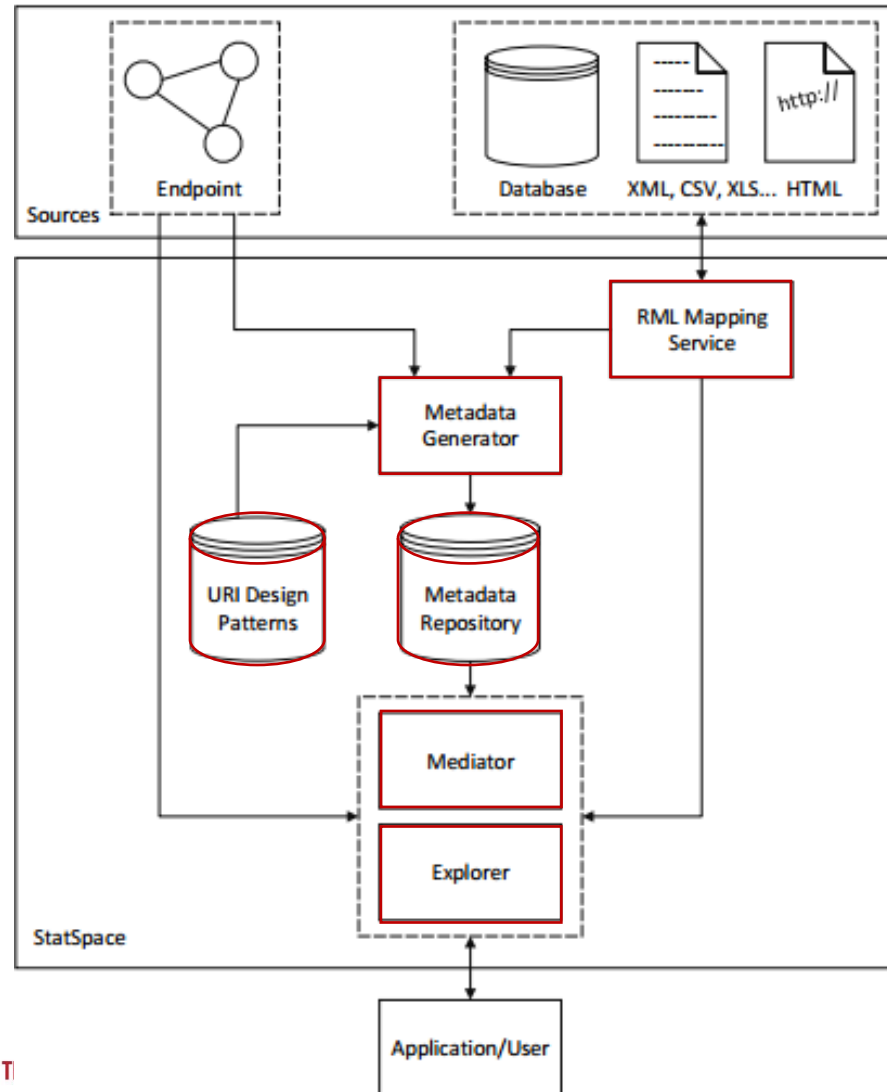


Metadata Description – Description of coreference relationships



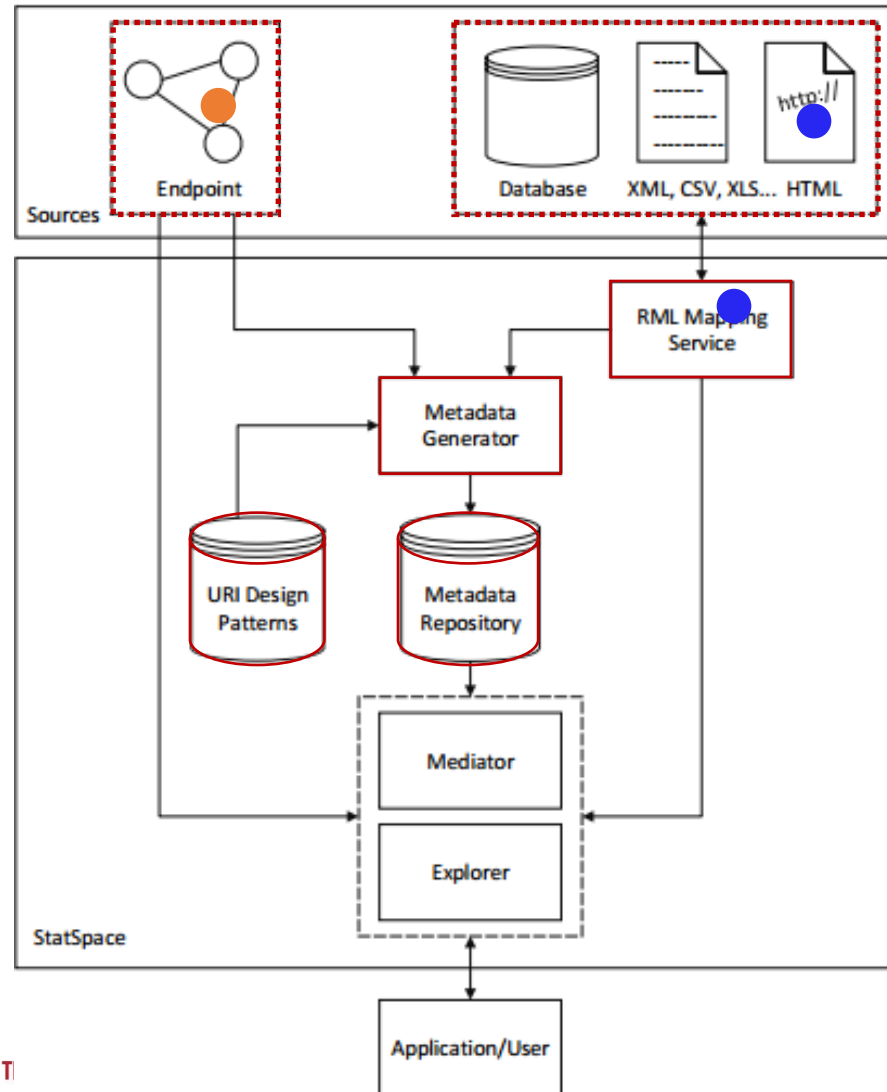
Architecture for Statistical Data Exploration and Integration

StatSpace Architecture



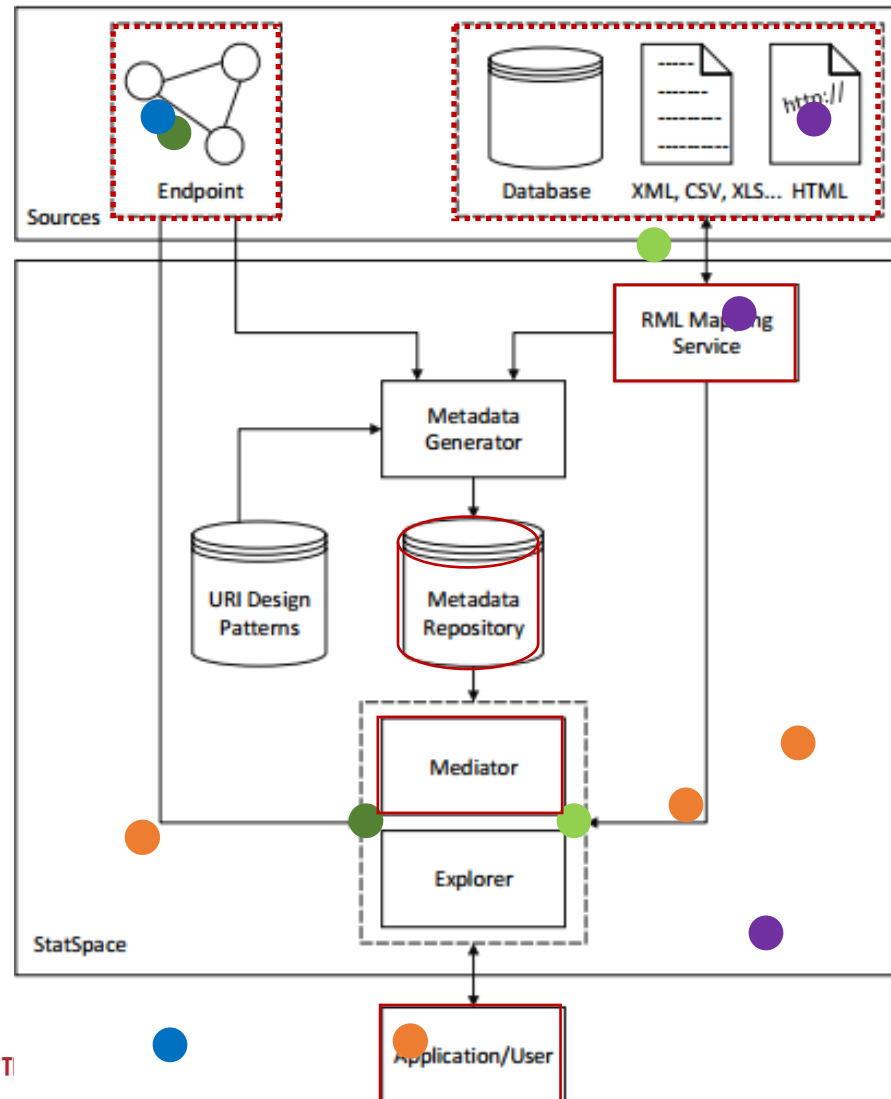
StatSpace Architecture

Metadata repository building



StatSpace Architecture

Mediator –
Cross-data set SPARQL querying



Mediator

Query Analysis

```
PREFIX qb:      <http://purl.org/linked-data/cube#>
PREFIX sdmxd:   <http://purl.org/linked-data/sdmx/2009/dimension#>
PREFIX sdmxm    <http://purl.org/linked-data/sdmx/2009/measure#>
PREFIX sdmxa:   <http://purl.org/linked-data/sdmx/2009/attribute#>
PREFIX dc:      <http://purl.org/dc/terms/>
```

Query Rewriting

```
SELECT * WHERE {
  ?ds dc:subject
    <http://statspace.linkedwidgets.org/codelist/cl_subject/SP.POP.TOTL>.
    ?o qb:dataSet ?ds.
    ?o sdmxm:obsValue ?obsValue.
    ?o sdmxd:refPeriod ?refPeriod.
    ?o sdmxd:refArea ?refArea.
    ?o sdmxa:unitMeasure ?unit.
```

Result Rewriting

```
Filter(?refArea=<
http://statspace.linkedwidgets.org/codelist/cl_area/UnitedKingdom>)) }
```

Result Integration

Mediator

Query
Analysis

```
SELECT * WHERE {  
  ?o qb:dataSet  
  <http://rdfdata.eionet.europa.eu/eurostat/data/demo_pjanbroad>.  
  ?o sdmxm:obsValue ?obsValue.  
  ?o sdmxd:timePeriod ?refPeriod.  
  ?o sdmxd:refArea ?refArea.  
  ?o sdmxd:freq <http://purl.org/linked-data/sdmx/2009/code#freq-A>.  
  ?o sdmxd:age <http://dd.eionet.europa.eu/vocabulary/eurostat/age/TOTAL>.  
  ?o sdmxd:sex <http://purl.org/linked-data/sdmx/2009/code#sex-T>.  
  FILTER(?refArea= <http://dd.eionet.europa.eu/vocabulary/eurostat/geo/UK>) }
```

Query
Rewriting

EEA data set query

Result
Rewriting

```
http://statspace.linkedwidgets.org/rml?rmlsource=http://statspace.linkedwidgets.org/mapping/wb.ttl&indicator=SP.POP.TOTL&refArea=GB
```

WB data set query

Result
Integration

```
http://statspace.linkedwidgets.org/rml?rmlsource=http://statspace.linkedwidgets.org/mapping/uk7.ttl
```

UK data set query

Mediator

Query
Analysis

- Rewrites each result based on co-reference relationships

Query
Rewriting

```
http://dd.eionet.europa.eu/vocabulary/eurostat/geo/UK =>  
http://statspace.linkedwidgets.org/codelist/cl_area/UnitedKingdom  
http://dd.eionet.europa.eu/vocabulary/worldbank/country/GB =>  
http://statspace.linkedwidgets.org/codelist/cl_area/UnitedKingdom
```

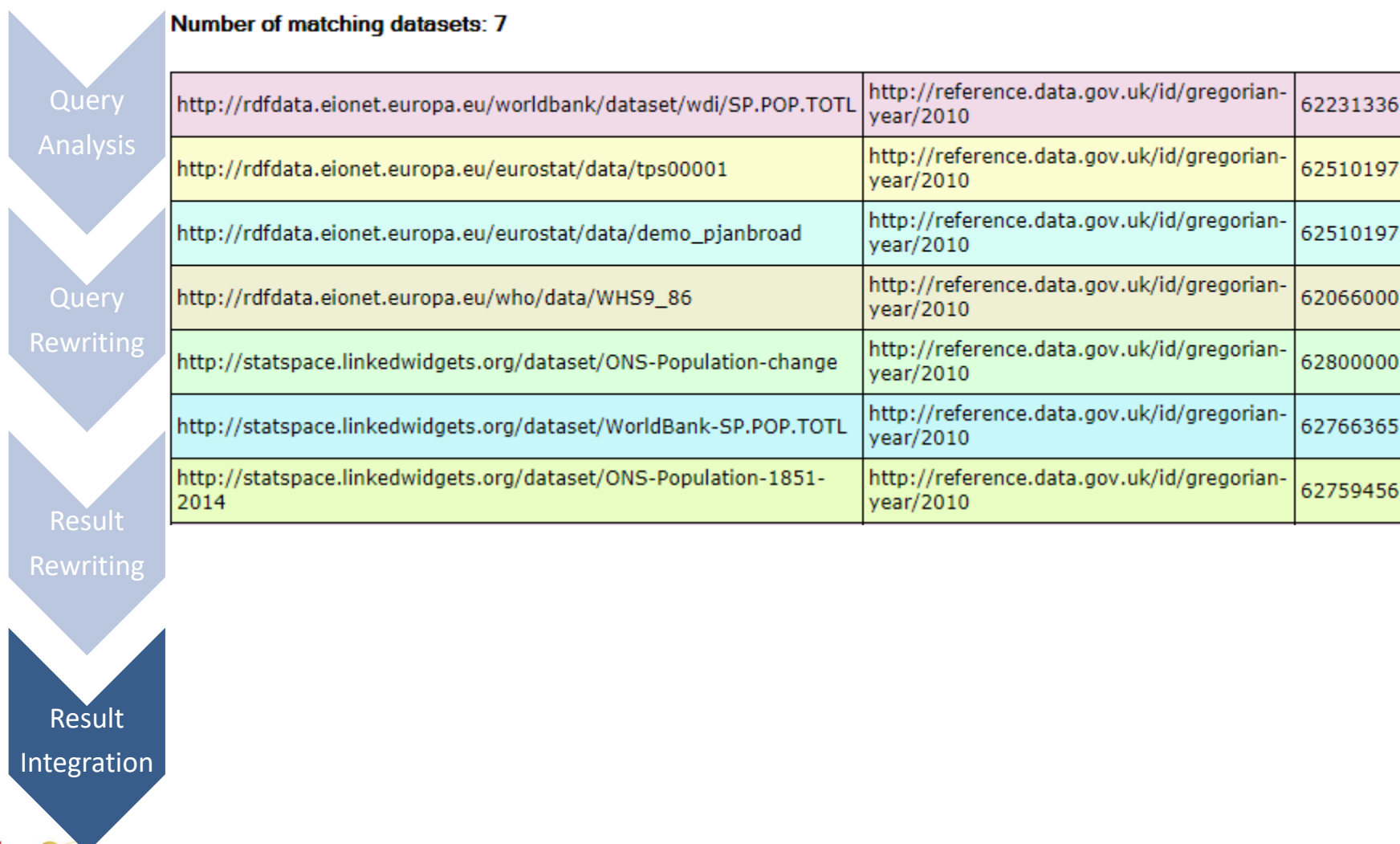
Result
Rewriting

- Consolidates different scales in results
 - EEA and WB data sets: absolute scale
 - UK data set: million scale => multiply one million

Result
Integration

- Integrates results

Mediator



Explorer

StatSpace Explorer

Search results:2151

- ☐ Narrow by Provider
- ☐ Narrow by Subject

Agricultural machinery, tractors

Provider: World Bank
Subject: AG.AGR.TRAC.NO

[Relatable datasets](#) [Visualization](#) [Source Metadata](#)

Fertilizer consumption (% of fertiliz...

Provider: World Bank
Subject: AG.CON.FERT.PT.ZS

[Relatable datasets](#) [Visualization](#) [Source Metadata](#)

Fertilizer consumption (kilograms per...

Provider: World Bank
Subject: AG.CON.FERT.ZS

[Relatable datasets](#) [Visualization](#) [Source Metadata](#)

Agricultural land (sq. km)

Provider: World Bank
Subject: AG.LND.AGRI.K2

[Relatable datasets](#) [Visualization](#) [Source Metadata](#)

Finding all statistical data sets about an area

StatSpace Explorer

Austria

Search

Search results: 1248

Narrow by Provider

World Bank	973
European Environment Age	128
European Union Open Data	119
Vienna OGD	23
United Nations Office on	4
European Environment Inf	1

Narrow by Subject

Agricultural machinery, tractors

Provider: World Bank
Subject: AG.AGR.TRAC.NO

[Relatable datasets](#) [Visualization](#) [Source Metadata](#)

Fertilizer consumption (kilograms per...

Provider: World Bank
Subject: AG.CON.FERT.ZS

[Relatable datasets](#) [Visualization](#) [Source Metadata](#)

Agricultural land (sq. km)

Provider: World Bank
Subject: AG.LND.AGRI.K2

[Relatable datasets](#) [Visualization](#) [Source Metadata](#)

Agricultural land (% of land area)

Provider: World Bank
Subject: AG.LND.AGRI.ZS

[Relatable datasets](#) [Visualization](#) [Source Metadata](#)

Identifying relatable data sets for a selected data set

GDP (current US\$)

Number of relatable datasets: 1792

Provider: World Bank
Subject: NY.GDP.MKTP.CD

[Visualization Source Metadata](#)

☐ Narrow by Provider

World Bank	1513
European Environment Age	141
European Union Open Data	119
Central Statistics Office	1
Vienna OGD	5
United Nations Office on	4
United Kingdom - Office	8
European Environment Inf	1

☐ Narrow by Subject

Gross nutrient balance on agricultura...

Provider: European Environment Agency (EEA)
Subject: AG.BAL.NUTR

[Comparison Visualization Source Metadata](#)

Labor force with intermediate educati...

Provider: World Bank
Subject: SL.TLF.INTM.MA.ZS

[Comparison Visualization Source Metadata](#)

Educational attainment, at least comp...

Provider: World Bank
Subject: SE.PRM.CUAT.ZS

[Comparison Visualization Source Metadata](#)

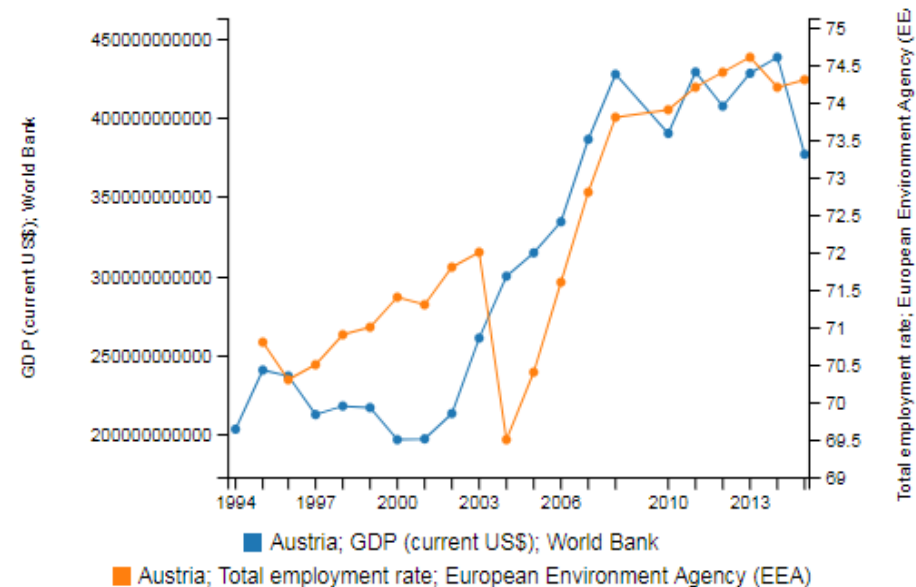
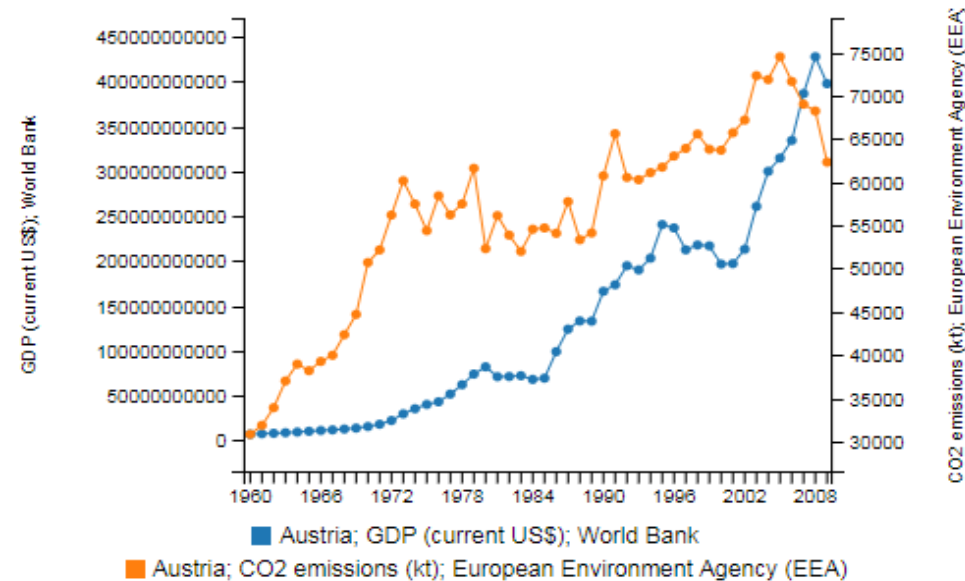
Time to export, documentary complianc...

Provider: World Bank
Subject: IC.EXP.TMDC

[Comparison Visualization Source Metadata](#)

Relatable data sets

Data visualization between multiple indicators



Comparing statistical data about different areas

StatSpace Explorer

Austria Germany GDP

Search

Search results: 91

[-] Narrow by Publisher

World Bank	82
European Environment Age	8
European Union Open Data	1

[+] Narrow by Subject

Trade in services (% of GDP)

Publisher: World Bank
Subject: BG.GSR.NFSV.GD.ZS

[Relatable datasets](#) [Visualization](#) [Source Metadata](#)

Foreign direct investment, net outflow...

Publisher: World Bank
Subject: BM.KLT.DINV.WD.GD.ZS

[Relatable datasets](#) [Visualization](#) [Source Metadata](#)

Current account balance (% of GDP)

Publisher: World Bank
Subject: BN.CAB.XOKA.GD.ZS

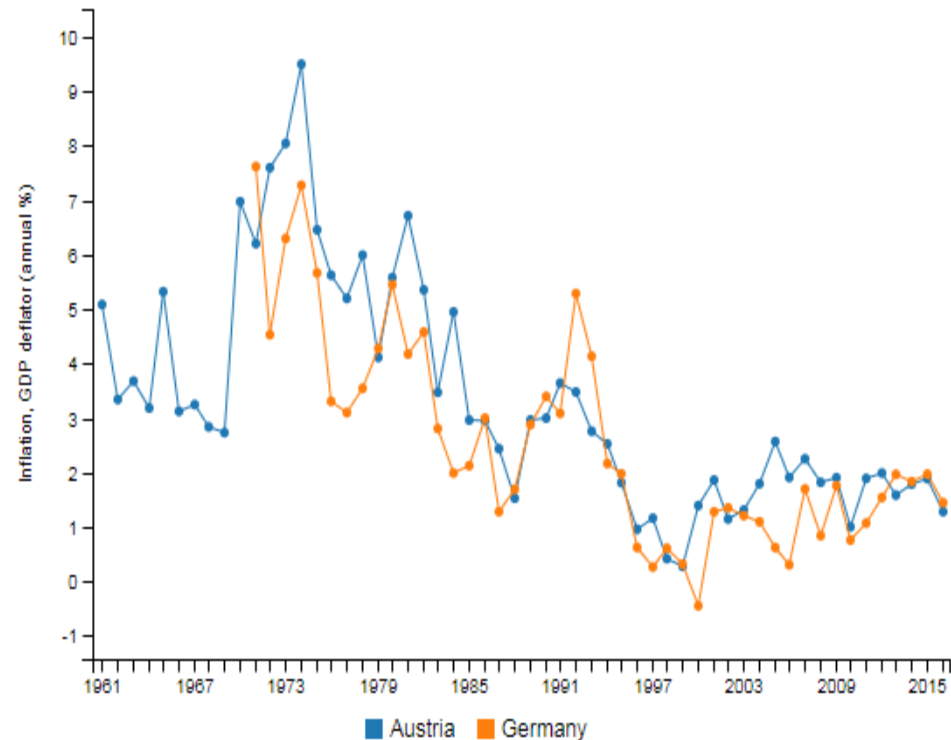
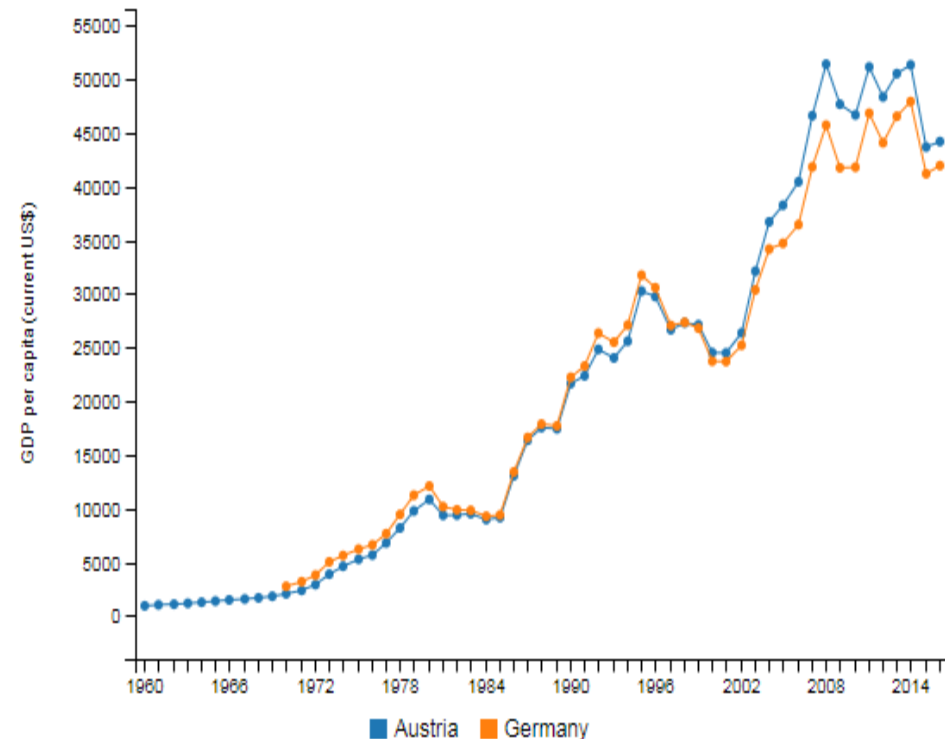
[Relatable datasets](#) [Visualization](#) [Source Metadata](#)

Foreign direct investment, net inflow...

Publisher: World Bank
Subject: BX.KLT.DINV.WD.GD.ZS

[Relatable datasets](#) [Visualization](#) [Source Metadata](#)

Comparing statistical data about different areas



Evaluation

Evaluation – Metadata Repository

- 601 descriptions of RDF data sets
- 1,459 descriptions of raw data sets

Source	Data format	Data sets used	Metadata descriptions	Metadata size (KB)
EUODP	RDF	151	151	826
EEA	RDF	147	147	3,331
CSO	RDF	61	61	413
ScotStat	RDF	23	23	428
ODC	RDF	3	3	268
VOGD	RDF	39	214	2,385
ONS	XLS	2	8	103
World Bank	JSON	1,451	1,451	22,045

Sources and numbers of data sets covered

Evaluation – Metadata Generator

- Goal: evaluates the correctness of the mappings
- Settings
 - 25 mappings for spatial dimension and 25 mappings for temporal dimension between the URIs of the European Environment Agency data and shared URIs
 - Five experts
- Result
 - All experts agree the correctness of 48 mappings
 - One mapping: lacks necessary information for evaluation
[2007^http://www.w3.org/2001/XMLSchema#int](http://www.w3.org/2001/XMLSchema#int) and
<http://reference.data.gov.uk/id/gregorian-year/2007>
 - One wrong mapping: due to error from Google's service
“Twente, Overijssel” => “Hof van Twente, Overijseel”
<http://dd.eionet.europa.eu/vocabulary/eurostat/geo/NL213> (label: Twente) and
http://statspace.linkedwidgets.org/codelist/cl_area/Netherlands/Overijssel/HofvanTwente

Evaluation - A comparison of research on statistical data integration

	Capadisli et al. [14]	Sabou et al. [11, 12]	Kämpgen et al. [26, 27]	Statspace
RQ1 { Original format	SDMX-XML	database	SPARQL	raw formats, SPARQL
RQ2 { Approach	warehousing	warehousing	warehousing	virtual integration
RQ2 { Requirements of data structure	same structure	–	same structure	relatable structures
RQ3 { Scale conversion	–	–	✓	✓
RQ3 { Uniform access	✗	✗	✗	✓

Conclusions and Future Work

Conclusions

How can users be enabled to explore and integrate heterogeneous statistical data sources?

- Architecture for statistical data exploration and integration
 - Up-to-date data to users
 - Uniform access to heterogeneous data sets
- Linked statistical data space: 1,800+ heterogeneous data sets from eight providers, <http://statspace.linkedwidgets.org/>

Research Questions

- RQ1. How can we address statistical data heterogeneity in terms of formats? **RDF, mapping language**
- RQ2. How can we establish interconnections between statistical data sets? **shared URIs, mapping algorithms**
- RQ3. How can we provide uniform access to heterogeneous statistical data sets? **metadata descriptions and mediator**

References

1. Maali et al. A dynamic faceted browser for data cube statistical data, Workshop on Using Open Data, 2012
2. Salas et al., Publishing Statistical Data on the Web, Conference on Semantic Computing, 2012
3. Klímek et al., Payola: Collaborative Linked Data Analysis and Visualization Framework, *ESWC 2013 Satellite Events*, 2013
4. Ermilov et al., Linked Open Data Statistics: Collection and Exploitation, *Conference on Knowledge Engineering and Semantic Web*, 2013
5. Hoefler et al., Linked Data Query Wizard: A Novel Interface for Accessing SPARQL Endpoints, LDOW, 2014
6. Kalampokis et al., Exploiting linked data cubes with opencube toolkit, ISWC Posters & Demos Track, 2014
7. Kämpgen et al., OLAP4LD: A Framework for Building Analysis Applications over Governmental Statistics, ESWC 2014 Satellite Events, 2014
8. Bayerl et al., Linked Data Integration based on the RDF Data Cube Vocabulary, *Conference on Web Intelligence, Mining and Semantics*, 2015

References

9. RDF Working Group, Resource Description Framework (RDF), <https://www.w3.org/RDF/>, 2014
10. Haase et al., A Comparison of RDF Query Languages, ISWC, 2004
11. Sabou et al., Tourmislod: A tourism linked data set, Semantic Web, 2013
12. Sabou et al., Linked Data for Cross Domain Decision-Making in Tourism, Conference on Information and Communication Technologies in Tourism, 2015
13. Kalampokis et al., Creating and utilizing linked open statistical data for the development of advanced analytics services, Semantics Statistics, 2014
14. Capadisli et al. Towards Linked Statistical Data Analysis, SemStat, 2013
15. Trinh et al., Linked Widgets: An Approach to Exploit Open Government Data, DATA conference, 2014

References

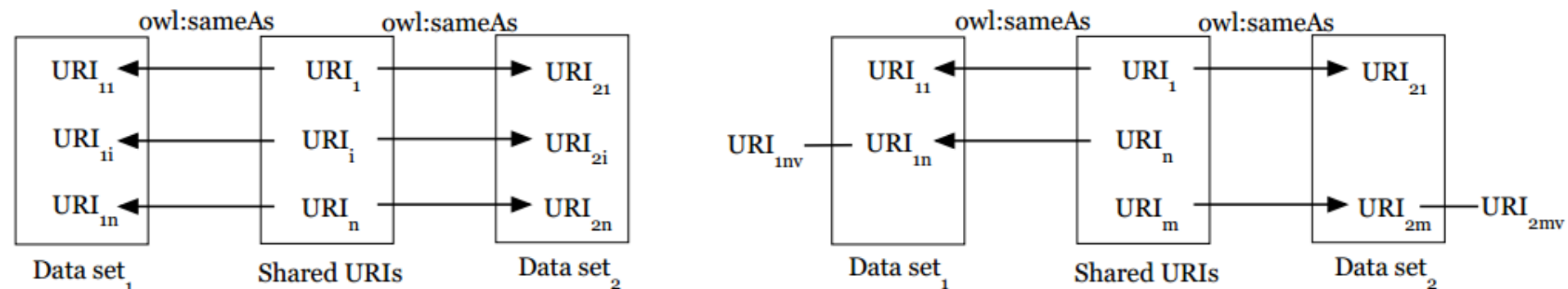
16. Langegger et al., XLWrap – Querying and Integrating Arbitrary Spreadsheets with SPARQL, ISWC, 2009
17. Connor et al., Mapping Master: A Flexible Approach for Mapping Spreadsheets to OWL, ISWC, 2010
18. Bischof et al., Mapping between RDF and XML with XSPARQL. *Journal on Data Semantics*, 2012
19. Bizer et al., D2R map-a database to RDF mapping language, *WWW – Poster Track*. 2003
20. Das et al., R2RML, <https://www.w3.org/TR/r2rml/>, 2012
21. Ghasemi et al., M2RML: Multidimensional to RDF Mapping Language, *Workshop on Database and Expert Systems Applications*, 2014
22. Dimou et al., RML: A Generic Language for Integrated RDF Mappings of Heterogeneous Data, LDOW, 2014
23. RML Processor. <https://github.com/RMLio/RML-Processor>

References

24. Hugh Glaser et al., Managing co-reference on the semantic web, LODW, 2009
25. Kai Schlegel et al., Balloon fusion: SPARQL rewriting based on unified co-reference information, *Conference on Data Engineering*, 2014
26. Kämpgen et al., Querying the Global Cube: Integration of Multidimensional Datasets from the Web, *Conference on Knowledge Engineering and Knowledge Management*, 2014
27. Benedikt Kämpgen, Flexible Integration and Efficient Analysis of Multidimensional Datasets from the Web, PhD Thesis, 2015
28. Mutlu et al., Automated Visualization Support for Linked Research Data, I-SEMANTICS Posters and Demos, 2013
29. Kotu et al., Predictive Analytics and Data Mining: Concepts and Practice with RapidMiner. Morgan Kaufmann, 2014
30. Saed Sayad. Data Exploration, http://chem-eng.utoronto.ca/~datamining/Presentations/Data_Exploration.pdf, 2010

BACKUP SLIDES

Relatable data structures

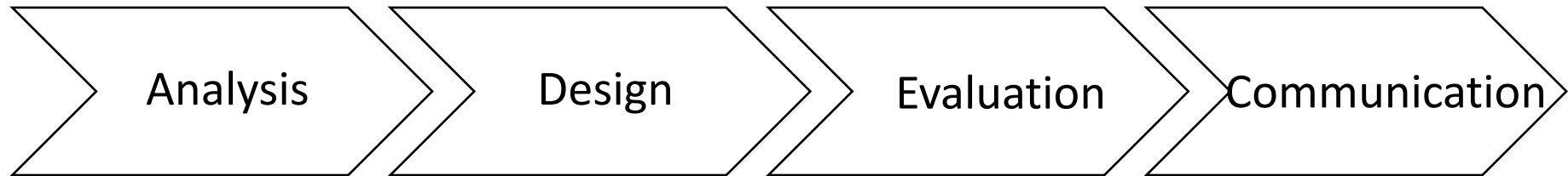


uk:country	uk:year	uk:value	uk:unit
uk:UnitedKingdom	uk:2013	64.1	uk:Million

wb:country	wb:year	wb:value	wb:unit
wb:GB	wb:2013	64128226	wb:AbsoluteScale

sdmxd:freq	sdmxd:timePeriod	sdmxd:refArea	sd-mxd:age	sdmxd:sex	sdmxm:obsValue
sdmx-code:freqA	2013	geo:UK	ag:TOTAL	sdmxcode:sex-T	63,905,297
sdmx-code:freqA	2013	geo:UK	ag:Y_LT15	sdmxcode:sex-T	11,260,549
sdmx-code:freqA	2013	geo:UK	ag:Y15-64	sdmxcode:sex-F	20,917,257

Design Science Research Methodology



Peppers et al., *A design science research methodology for information systems research*,
Journal of management information systems, 2007

Shared URIs – Overview

- URIs for Components
 - 11 URIs for dimensions, e.g., sdmxd:refArea, sdmxd:sex, sdmxd:freq, sdmxd:occupation, sdmx:civilStatus, etc.
 - 1 URI for measure (generic measure): sdmxm:obsValue
 - 1 URI for attribute (unit of measure): sdmxa:unitMeasure
- URIs for Subjects
 - *Topic.General Subject.Specific Subject.Extensions* (e.g. *SP.POP.TOTL.FE*)
 - 55 codes for Topics, 198 codes for General topics, 629 codes for Specific Subjects, 147 codes for Extensions
- URIs for Units: 43 URIs, e.g.,
http://statspace.linkedwidgets.org/codelist/cl_unitMeasure/P6,
refers to **people in million** (10^6)

Shared URIs – Code lists of Components

Code list	URI design patterns	Number of URIs	Hierarchy Support
CL_Area	1	Unlimited	Yes
CL_Period	11	Unlimited	Yes
CL_Economic_Activity	1	996	Yes
CL_Age	1	209	Yes
CL_Education_Level	1	9	No
CL_COICOP	1	230	Yes
CL_COFOP	1	188	Yes
CL_COPP	1	51	Yes
CL_COPNI	1	65	Yes
CL_Occupation	1	619	Yes
CL_Currency	1	180	No
CL_Civil_Status	1	8	No
CL_Frequency	1	9	No
CL_Sex	1	5	Yes
CL_Unit_Measure	1	43	No

Characteristics of code lists

Shared URIs – Code lists of Components

The *expenditure dimension* consists of four code lists, i.e.,

- classification of individual consumption by purpose (COICOP): food, clothing, etc.
- classification of functions of government (COFOG): defense, health, etc.
- classification of purposes of non-profit institutions serving households (COPNI): R&D for health, R&D for education, etc.
- classification of outlays of producers by purpose (COPP): sales promotion, maintenance, etc.

Evaluation – Shared URIs

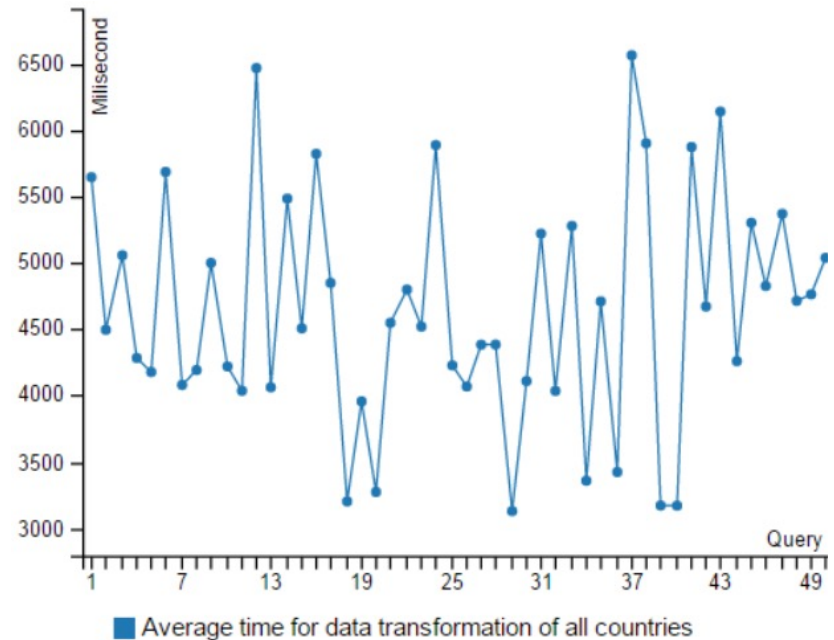
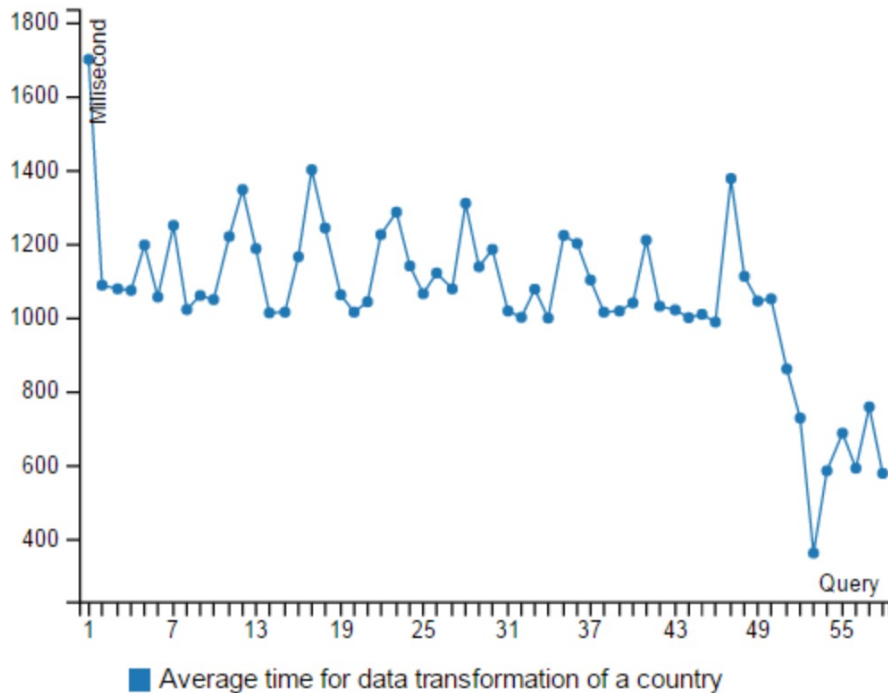
Source	Area	Time	Age	Sex	Fre- quency	Occu- pation
EUODP	31	12	0	0	0	0
EEA	3,114	609	132	0	3	0
ScotStat	16,718	167	83	3	0	0
CSO	4,806	1	40	3	0	11
ODC	33,286	167	0	0	4	0
VOGD	2,542	823	113	3	0	0
World Bank	304	57	–	–	–	–
URI Design Patterns	Unlim- ited	Unlim- ited	209	5	9	619

A comparison of size of code lists for six popular dimensions

Evaluation – RML Mapping Service

- Test environment
 - Windows 7 Enterprise 64 bit, Ram - 8.00 GB of DDR3, Processor - Intel Core i5-3470, CPU@3.20 GHz
 - Internet connection speed: around 800 Mbps for download and 900 Mbps for upload
- Number of queries
 - 50 queries to transform data of a single country from the WB
 - 8 queries to transform data from the UK data source
 - 50 queries to transform data of all countries from the WB

Evaluation – RML Mapping Service



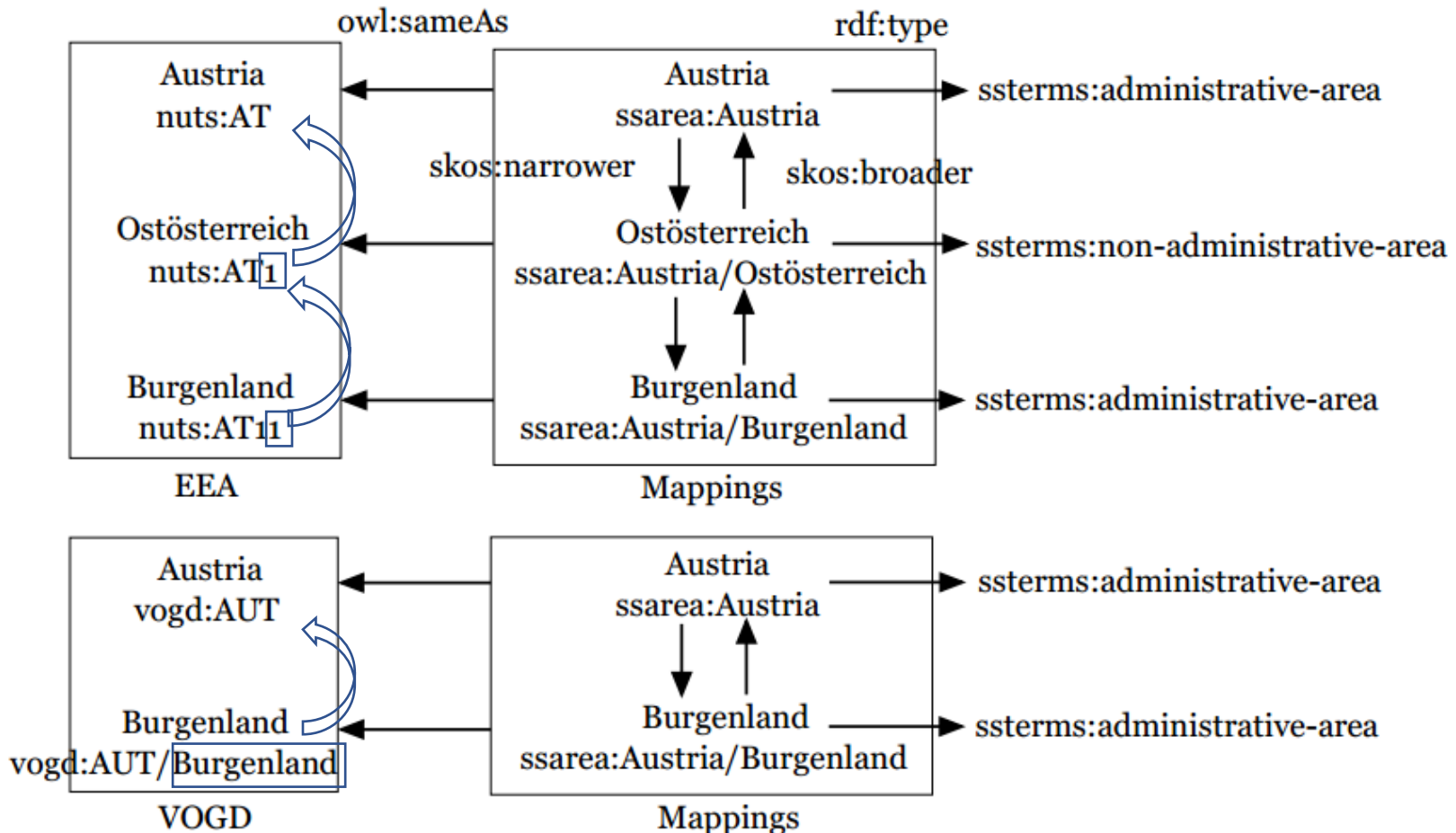
Limitations

- Data collection methodology is not modelled
- Territorial changes and differences in fiscal years are not considered
- Do not have participation of data publishers/experts to valid mappings between URIs used in data sources and shared URIs
- Mediator's implementation needs to be improved

Spatial Dimension Mapping Algorithm

- Ambiguity: *Vienna – Austria, Vienna – USA*
 - Orders input areas in the ascending order of geographical levels
 - Adds label of the broader area (e.g., Austria) to the queries of its narrower areas (e.g., Vienna)
- Direction-based regions: Ostösterreich - Eastern Austria
 - Assigns URI based on URI of its broader area and its label
 - Sets type “non-administrative area”

Spatial Dimension Mapping Algorithm - Example

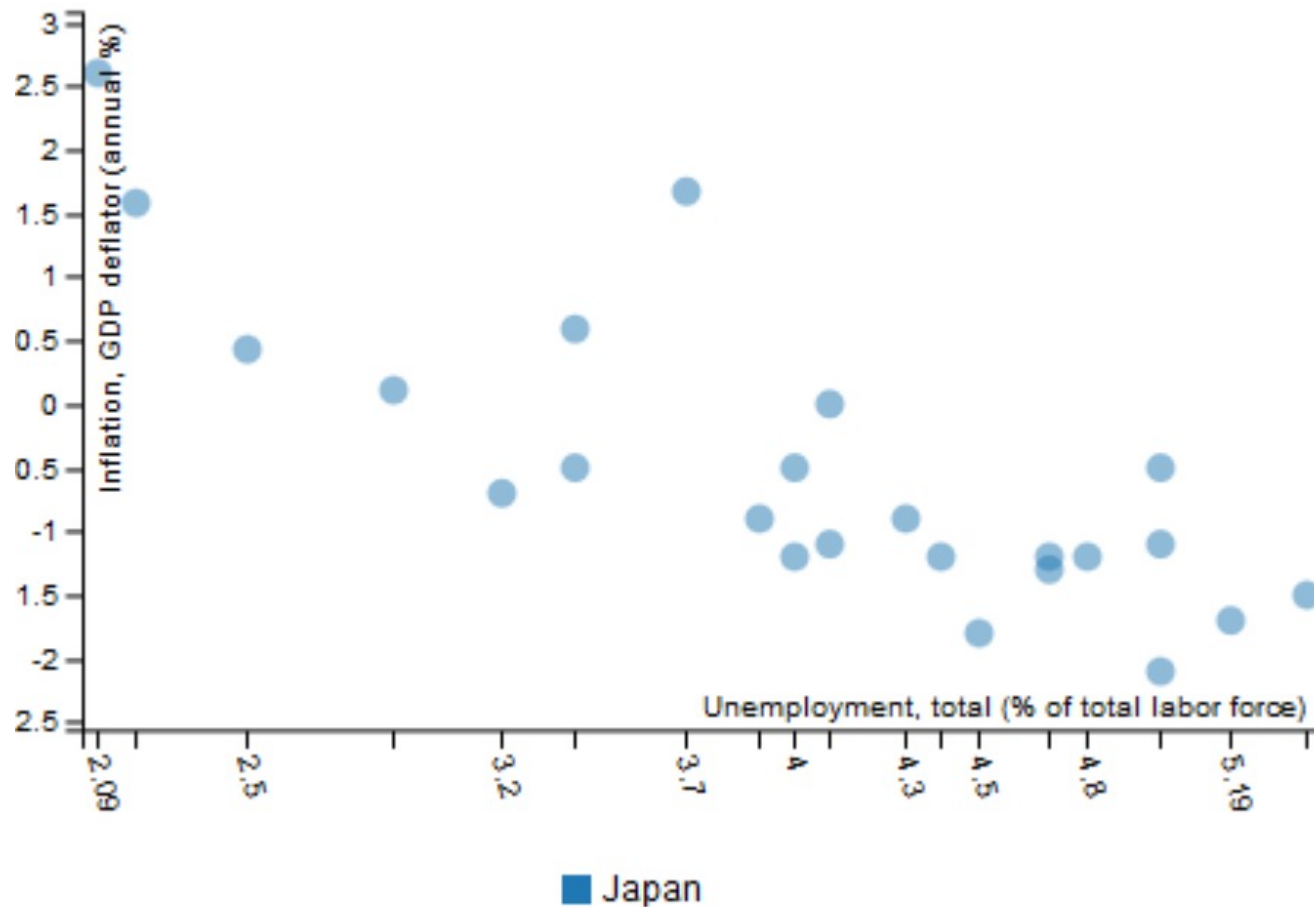


nuts: <http://dd.eionet.europa.eu/vocabulary/common/nuts/>

vogd: <http://ogd.ifs.tuwien.ac.at/vienna/geo/>

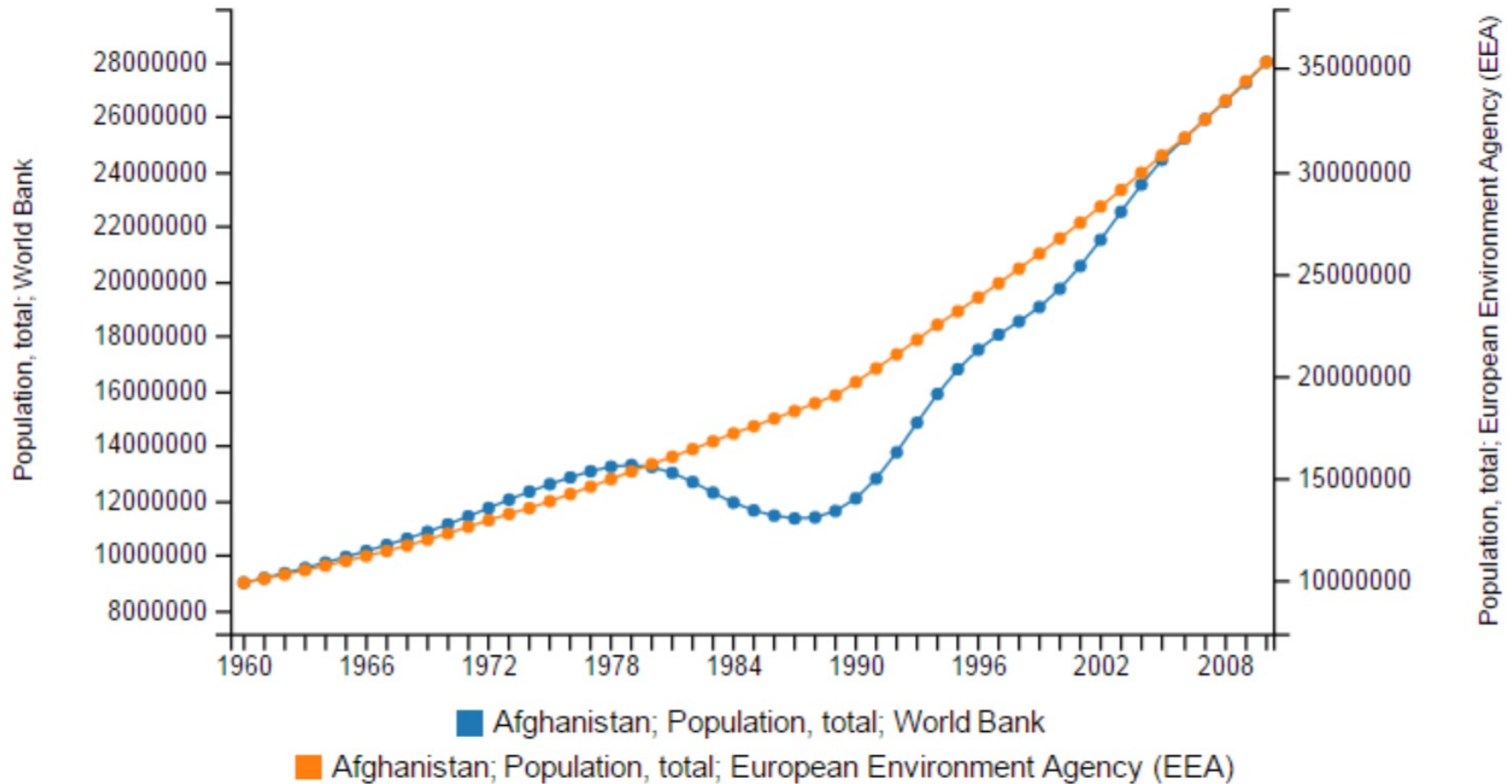
ssarea: http://statspace.linkedwidgets.org/codelist/cl_area/

Use Case – Scatter plot for Correlation Mining

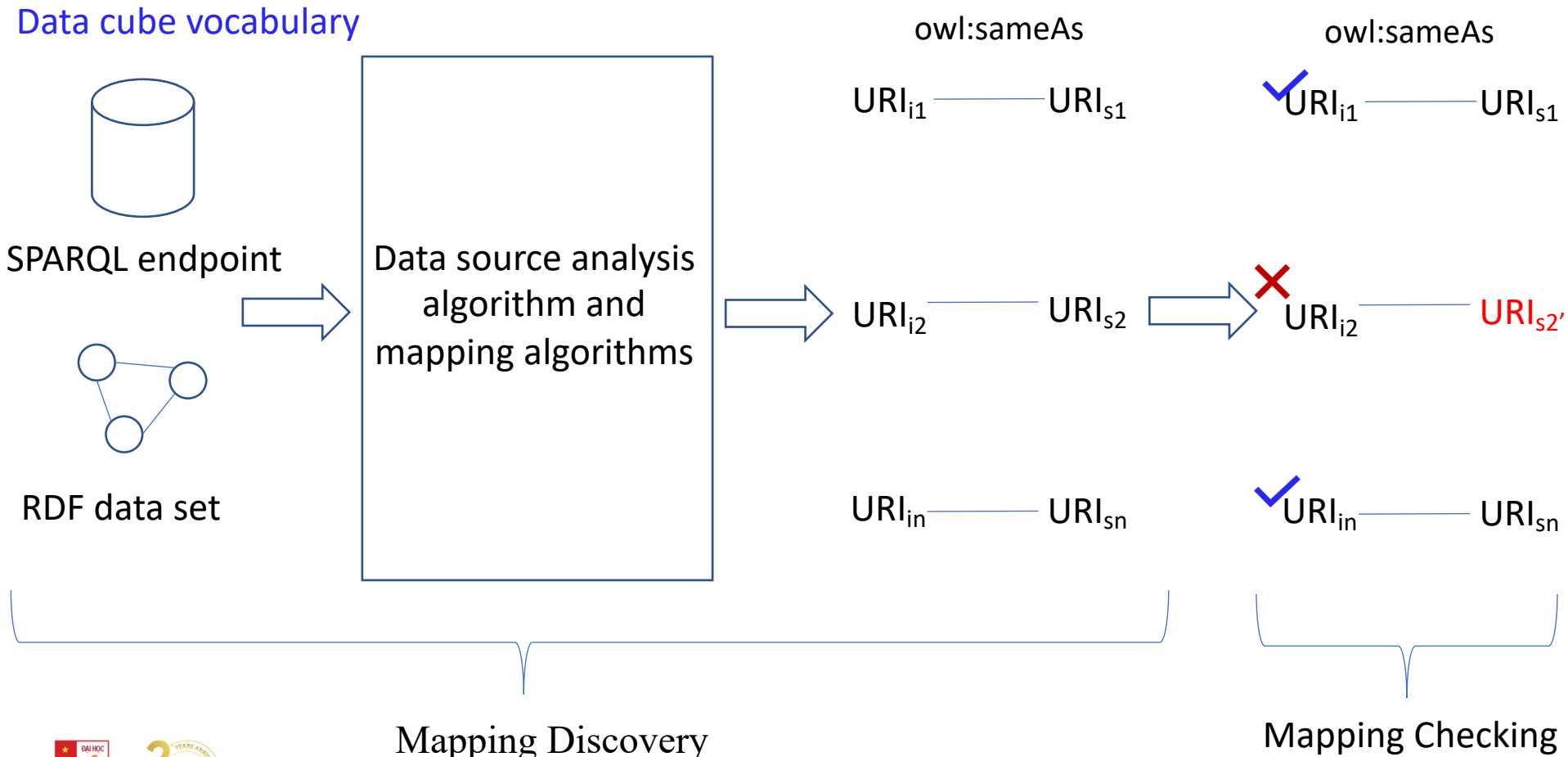


Relationship between Inflation and Unemployment indicators for Japan

Use Case – Showing Out-of-date Data



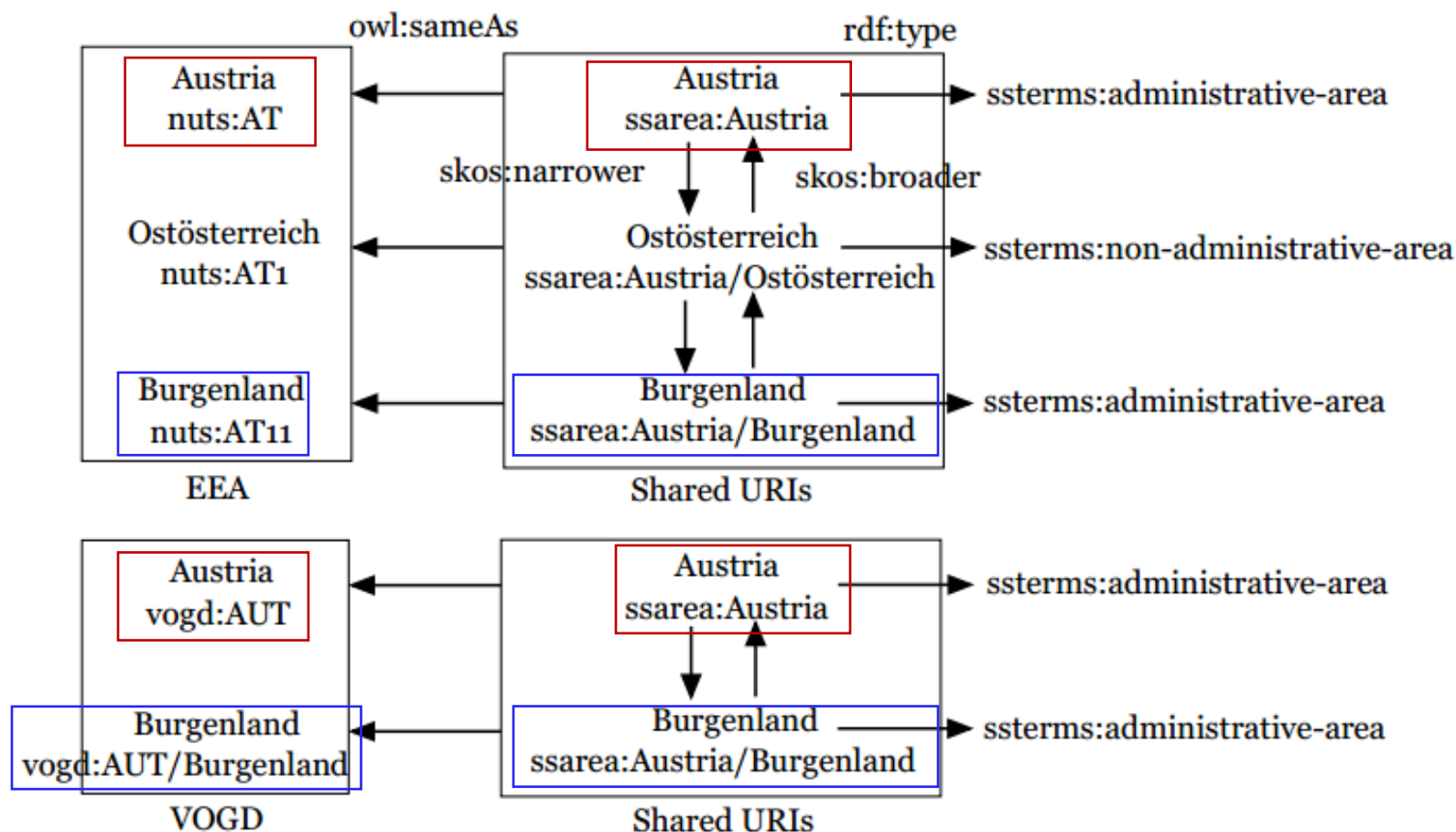
Mapping Generation



Mapping Generation

- Data source analysis algorithm
 - Input: a SPARQL endpoint or an RDF data set following Data cube vocabulary
 - Output: A list of data sets, each contains label, components, and their values
- Mapping algorithms
 - Eleven mapping algorithms for eleven dimensions
 - One mapping algorithm for the unit attribute
 - Mapping algorithm for spatial dimension: relies on Google's geocoding service
 - Mapping algorithms for other components: rely on patterns and keywords

Spatial Dimension Mapping Algorithm - Example

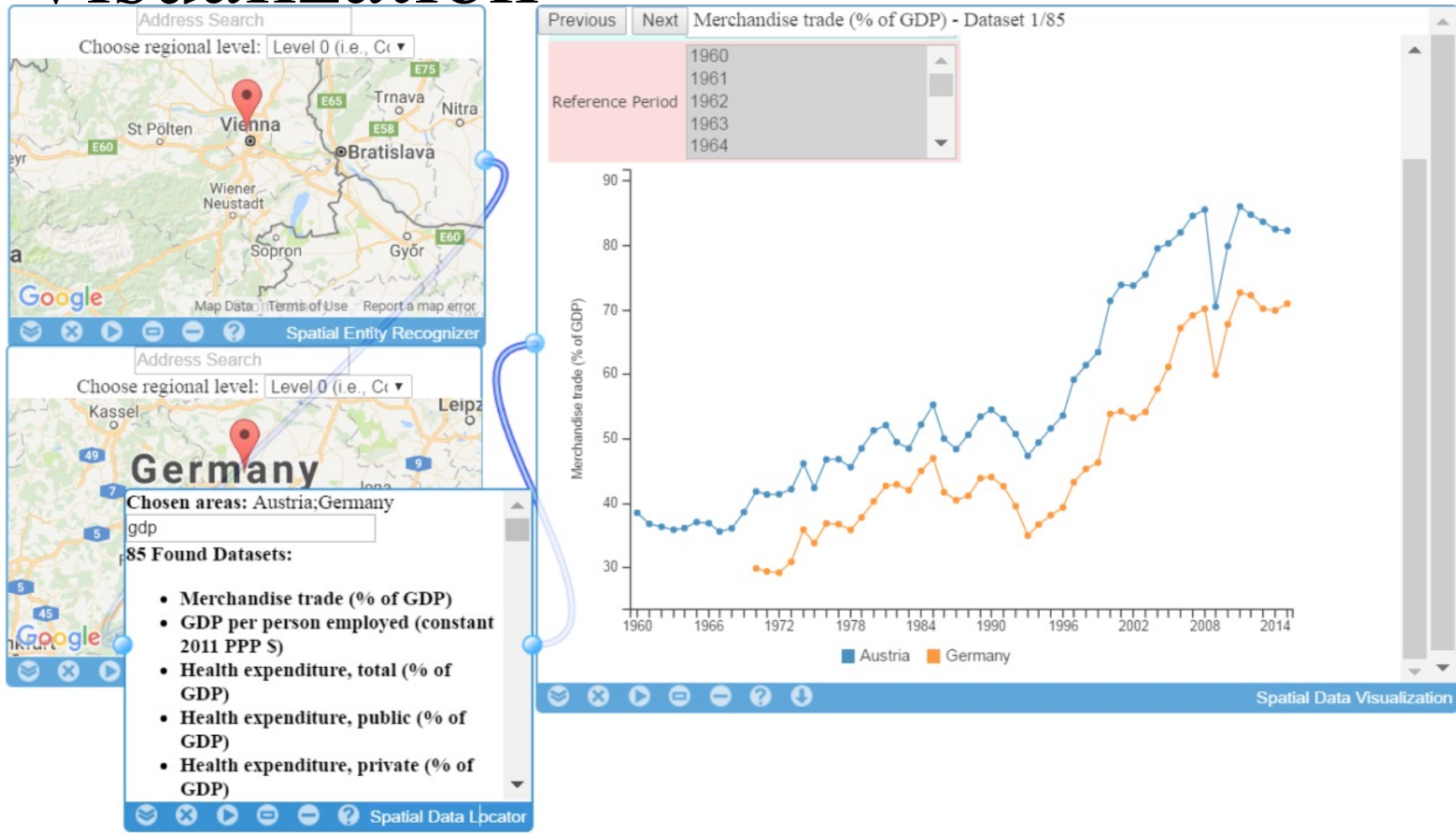


nuts: <http://dd.eionet.europa.eu/vocabulary/common/nuts/>

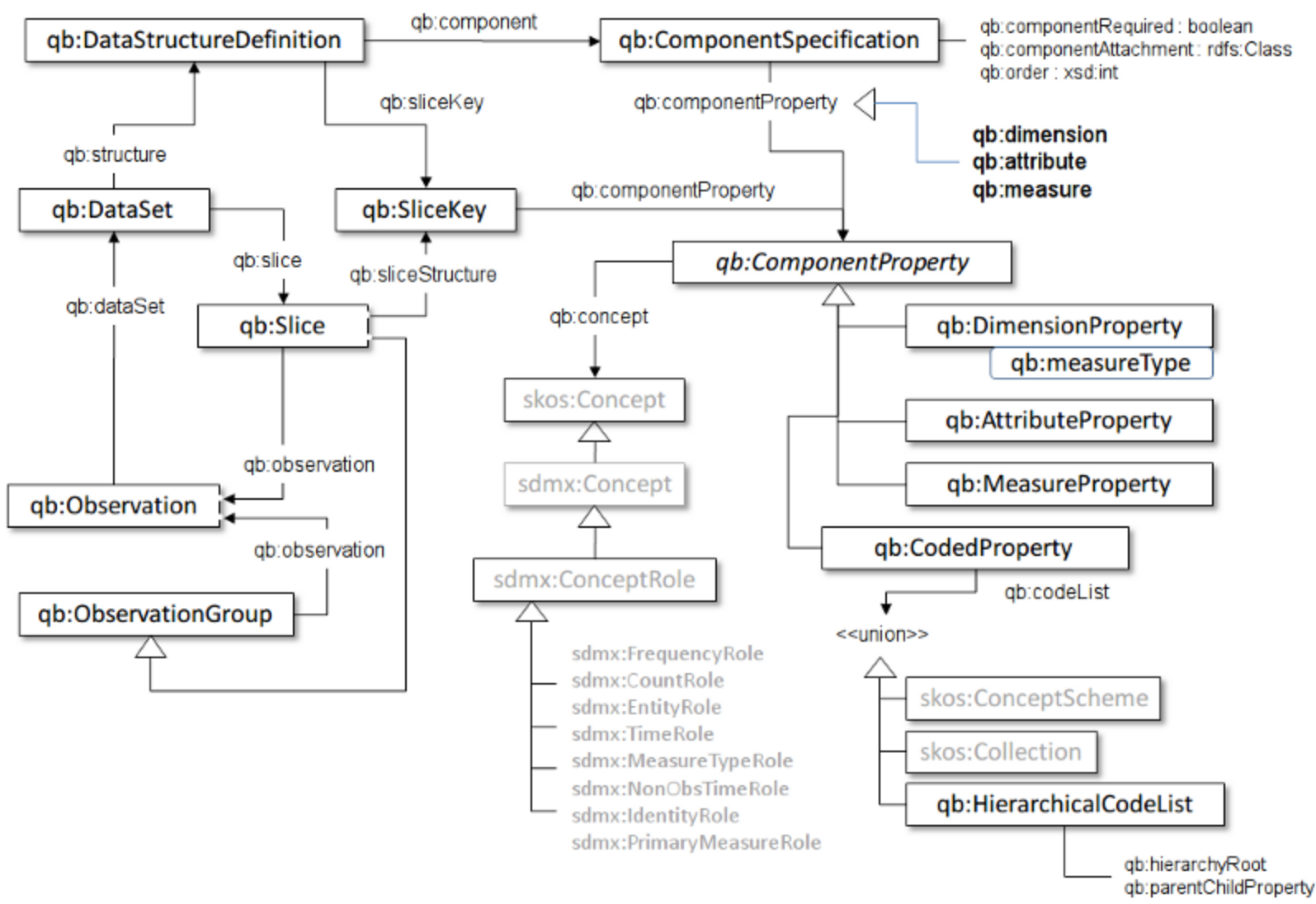
vogd: <http://ogd.ifs.tuwien.ac.at/vienna/geo/>

ssarea: http://statspace.linkedwidgets.org/codelist/cl_area/

Use Case – Spatial Data Visualization



Data Cube Vocabulary



Data Cube Vocabulary - Example

